

# CafeF–FinABSA Validation Report

Báo cáo sinh t ng t  d  li u và k t qu  trong repository.

L u ý: Các k t lu n h c thu t c n    c ki m tra l i sau khi model output ho n t t.

## 1. Executive summary

S  d ng input: 208.

S  d ng prediction: 208.

S  nh n ch a parse    c: 0.

K t qu  ch nh    c thi t k  theo next-trading-day return và abnormal return d a tr n market model.   y l  ki m   nh li n h  ngoài m u, kh ng ph i b ng ch ng nh n qu  hay khu n ngh     u t .

## 2. Prediction validation

```
{
  "input_rows": 208,
  "prediction_rows": 208,
  "missing_predictions": 0,
  "unexpected_predictions": 0,
  "duplicate_input_ids": 0,
  "duplicate_prediction_ids": 0,
  "unknown_labels": 0,
  "label_counts": {
    "neutral": 180,
    "positive": 18,
    "negative": 10
  }
}
```

## 3. Event-study summary

| label    | events   | mean_sentiment | mean_return | mean_abnormal_return | median_abnormal_return | mean_article_count |
|----------|----------|----------------|-------------|----------------------|------------------------|--------------------|
| negative | 10.0000  | -1.0000        | -1.4767     | -0.3665              | -1.2829                | 1.0000             |
| neutral  | 172.0000 | 0.0000         | -0.8231     | 0.1105               | 0.1322                 | 1.3081             |
| positive | 18.0000  | 1.0000         | 0.3498      | 1.1637               | 0.9237                 | 1.3333             |

## 4. Panel regression

| Unnamed: 0       | coef    | std_err | p_value |
|------------------|---------|---------|---------|
| Intercept        | -0.2803 | 1.1866  | 0.8133  |
| C(ticker)[T.AMD] | -5.2975 | 14.2565 | 0.7102  |
| C(ticker)[T.BID] | 1.6386  | 1.7063  | 0.3369  |
| C(ticker)[T.CEN] | 2.4069  | 2.2224  | 0.2788  |
| C(ticker)[T.CTG] | 1.1301  | 2.3490  | 0.6304  |
| C(ticker)[T.DGC] | 0.6507  | 2.9762  | 0.8269  |

| Unnamed: 0       | coef    | std_err | p_value |
|------------------|---------|---------|---------|
| C(ticker)[T.DIG] | -0.9622 | 3.6341  | 0.7912  |
| C(ticker)[T.DNP] | -1.3602 | 10.3561 | 0.8955  |
| C(ticker)[T.DXG] | -7.7002 | 4.2263  | 0.0685  |
| C(ticker)[T.EIB] | 1.5188  | 4.5640  | 0.7393  |
| C(ticker)[T.FTS] | 0.1322  | 1.8762  | 0.9438  |
| C(ticker)[T.GAS] | -0.9506 | 3.6477  | 0.7944  |
| C(ticker)[T.GEX] | 0.2141  | 2.0003  | 0.9148  |
| C(ticker)[T.HAG] | -2.6198 | 1.9800  | 0.1858  |
| C(ticker)[T.HBC] | -0.4332 | 6.6026  | 0.9477  |
| C(ticker)[T.HCM] | -0.3415 | 2.8158  | 0.9035  |
| C(ticker)[T.HDB] | 2.0872  | 36.5690 | 0.9545  |
| C(ticker)[T.HDC] | -5.9374 | 2.4916  | 0.0172  |
| C(ticker)[T.HPG] | -0.6630 | 1.5042  | 0.6594  |
| C(ticker)[T.HSG] | -7.5340 | 10.1227 | 0.4567  |
| C(ticker)[T.ICT] | -0.7350 | 2.1511  | 0.7326  |
| C(ticker)[T.ITA] | -2.2455 | 1.3423  | 0.0943  |
| C(ticker)[T.KBC] | -1.8416 | 1.3364  | 0.1682  |
| C(ticker)[T.LHG] | -3.5513 | 4.9816  | 0.4759  |
| C(ticker)[T.LPB] | 2.0555  | 2.1186  | 0.3319  |
| C(ticker)[T.MBS] | -0.5023 | 1.8206  | 0.7826  |
| C(ticker)[T.MSH] | 0.3562  | 5.4325  | 0.9477  |
| C(ticker)[T.MWG] | -3.4896 | 1.3126  | 0.0078  |
| C(ticker)[T.NKG] | -7.4478 | 1.7070  | 0.0000  |
| C(ticker)[T.NVL] | -0.1798 | 1.6682  | 0.9142  |
| C(ticker)[T.OCB] | -0.6911 | 1.3638  | 0.6123  |
| C(ticker)[T.PNJ] | -2.7945 | 2.1092  | 0.1852  |
| C(ticker)[T.POW] | -0.3177 | 1.5848  | 0.8411  |
| C(ticker)[T.PVD] | 4.3167  | 12.1104 | 0.7215  |
| C(ticker)[T.REE] | -0.5713 | 1.6876  | 0.7349  |
| C(ticker)[T.SAB] | 1.5371  | 2.1983  | 0.4844  |
| C(ticker)[T.SHB] | 0.7916  | 1.3222  | 0.5494  |
| C(ticker)[T.SSI] | -0.0785 | 1.5113  | 0.9586  |
| C(ticker)[T.STB] | 1.4646  | 9.1620  | 0.8730  |
| C(ticker)[T.TCB] | -0.4559 | 2.3430  | 0.8457  |
| C(ticker)[T.TGG] | -2.0822 | 1.5986  | 0.1927  |
| C(ticker)[T.TNI] | -1.3341 | 1.6645  | 0.4228  |

| Unnamed: 0                                         | coef    | std_err | p_value |
|----------------------------------------------------|---------|---------|---------|
| C(ticker)[T.VCB]                                   | -4.3109 | 1.9044  | 0.0236  |
| C(ticker)[T.VIC]                                   | 0.4284  | 2.4304  | 0.8601  |
| C(ticker)[T.VIX]                                   | -2.0476 | 2.6596  | 0.4414  |
| C(ticker)[T.VND]                                   | -2.0885 | 1.4334  | 0.1451  |
| C(ticker)[T.VNM]                                   | -2.2090 | 1.5033  | 0.1417  |
| C(ticker)[T.VPB]                                   | 0.8495  | 6.0108  | 0.8876  |
| C(ticker)[T.VPS]                                   | -2.3774 | 2.1407  | 0.2667  |
| C(ticker)[T.VRE]                                   | 1.1187  | 6.2108  | 0.8571  |
| C(ticker)[T.VVS]                                   | 9.1739  | 8.1399  | 0.2597  |
| C(target_date)[T.Timestamp('2022-10-04 00:00:00')] | -1.7659 | 2.5124  | 0.4822  |
| C(target_date)[T.Timestamp('2022-10-05 00:00:00')] | 0.7433  | 1.9522  | 0.7034  |
| C(target_date)[T.Timestamp('2022-10-06 00:00:00')] | -1.7639 | 2.0151  | 0.3814  |
| C(target_date)[T.Timestamp('2022-10-07 00:00:00')] | -0.5156 | 1.1605  | 0.6569  |
| C(target_date)[T.Timestamp('2022-10-10 00:00:00')] | 1.9149  | 1.0443  | 0.0667  |
| C(target_date)[T.Timestamp('2022-10-11 00:00:00')] | -3.7897 | 1.3530  | 0.0051  |
| C(target_date)[T.Timestamp('2022-10-12 00:00:00')] | -0.6759 | 1.7647  | 0.7017  |
| C(target_date)[T.Timestamp('2022-10-13 00:00:00')] | 0.6902  | 1.3988  | 0.6217  |
| C(target_date)[T.Timestamp('2022-10-14 00:00:00')] | 4.6447  | 1.6437  | 0.0047  |
| C(target_date)[T.Timestamp('2022-10-17 00:00:00')] | 2.0972  | 0.9322  | 0.0245  |
| C(target_date)[T.Timestamp('2022-10-18 00:00:00')] | 2.0014  | 2.0668  | 0.3329  |
| C(target_date)[T.Timestamp('2022-10-19 00:00:00')] | -1.2883 | 2.4959  | 0.6057  |
| C(target_date)[T.Timestamp('2022-10-20 00:00:00')] | -2.8164 | 3.3729  | 0.4037  |
| C(target_date)[T.Timestamp('2022-10-21 00:00:00')] | -1.7624 | 1.1935  | 0.1398  |
| C(target_date)[T.Timestamp('2022-10-24 00:00:00')] | -0.3984 | 1.2502  | 0.7500  |
| C(target_date)[T.Timestamp('2022-10-25 00:00:00')] | 0.5612  | 1.5384  | 0.7153  |
| C(target_date)[T.Timestamp('2022-10-26 00:00:00')] | -0.5799 | 1.1070  | 0.6003  |
| C(target_date)[T.Timestamp('2022-10-27 00:00:00')] | 1.0854  | 1.2191  | 0.3733  |
| C(target_date)[T.Timestamp('2022-10-28 00:00:00')] | -0.2240 | 1.4596  | 0.8780  |
| C(target_date)[T.Timestamp('2022-10-31 00:00:00')] | -0.3527 | 1.1641  | 0.7619  |
| sentiment_score                                    | 0.0000  | 1.2603  | 1.0000  |
| negative_share                                     | -0.4822 | 1.7758  | 0.7860  |
| positive_share                                     | 0.9243  | 1.7058  | 0.5879  |
| log_article_count                                  | 1.4728  | 1.2080  | 0.2228  |

## 5. Data coverage

S■ ngày aggregate: 21.

S■ ticker có market model: 54.

| ticker | alpha   | beta    | n_estimation | model_status |
|--------|---------|---------|--------------|--------------|
| ACB    | 0.0288  | 0.9434  | 105.0000     | market_model |
| AMD    | -0.6832 | 1.4887  | 105.0000     | market_model |
| BID    | 0.1454  | 1.4112  | 105.0000     | market_model |
| BIG    | -0.3924 | 0.3397  | 105.0000     | market_model |
| CEN    | -0.2903 | 1.1666  | 105.0000     | market_model |
| CTG    | 0.0958  | 1.4386  | 105.0000     | market_model |
| DGC    | -0.0688 | 1.5952  | 105.0000     | market_model |
| DIG    | -0.2028 | 1.7626  | 105.0000     | market_model |
| DNP    | 0.0483  | -0.0055 | 99.0000      | market_model |
| DXG    | -0.2254 | 1.7109  | 105.0000     | market_model |
| EIB    | 0.1702  | 0.1483  | 105.0000     | market_model |
| FLC    | -0.7855 | 1.5124  | 89.0000      | market_model |
| FTS    | 0.1854  | 1.8623  | 105.0000     | market_model |
| GAS    | 0.2293  | 1.1169  | 105.0000     | market_model |
| GEX    | -0.0490 | 1.7771  | 105.0000     | market_model |
| HAG    | 0.4616  | 1.1152  | 105.0000     | market_model |
| HBC    | 0.0116  | 1.5022  | 105.0000     | market_model |
| HCM    | 0.2349  | 1.6810  | 105.0000     | market_model |
| HDB    | 0.1528  | 1.1120  | 105.0000     | market_model |
| HDC    | -0.0407 | 1.6777  | 105.0000     | market_model |
| HPG    | -0.2097 | 1.0532  | 105.0000     | market_model |
| HSG    | -0.1268 | 1.4914  | 105.0000     | market_model |
| ICT    | 0.1137  | 0.4238  | 105.0000     | market_model |
| ITA    | -0.5754 | 1.5134  | 105.0000     | market_model |
| KBC    | 0.1289  | 1.3851  | 105.0000     | market_model |
| LHG    | -0.1524 | 1.5908  | 105.0000     | market_model |
| LPB    | -0.0106 | 1.4453  | 105.0000     | market_model |
| MBS    | 0.1213  | 2.1671  | 105.0000     | market_model |
| MSH    | -0.2744 | 1.2546  | 105.0000     | market_model |
| MWG    | 0.0632  | 1.2326  | 105.0000     | market_model |

## 6. Figures

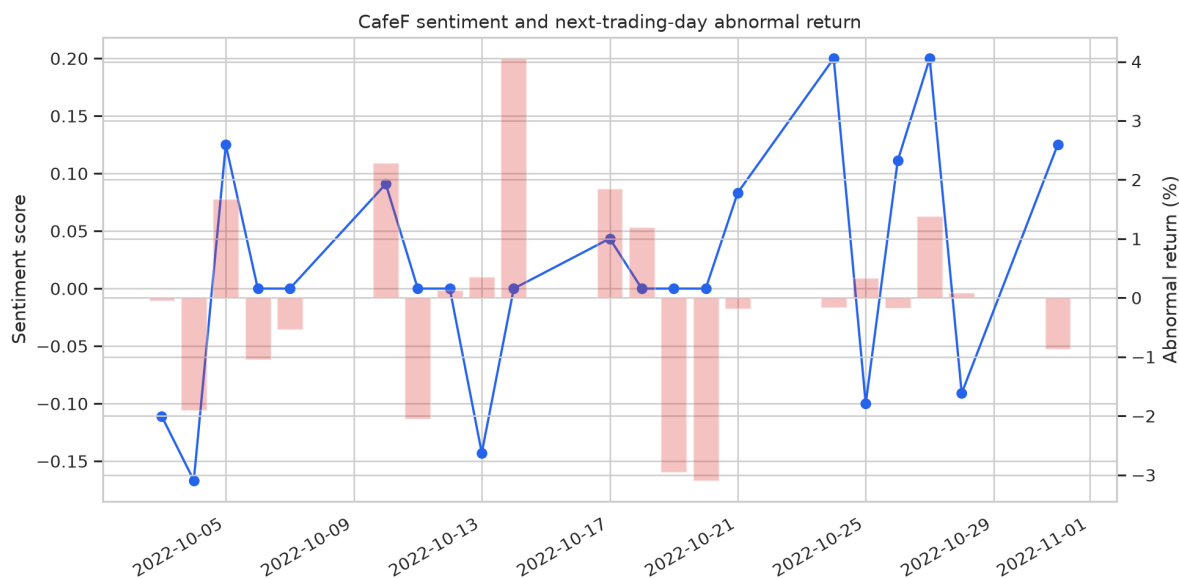


Figure 1. Sentiment versus abnormal return

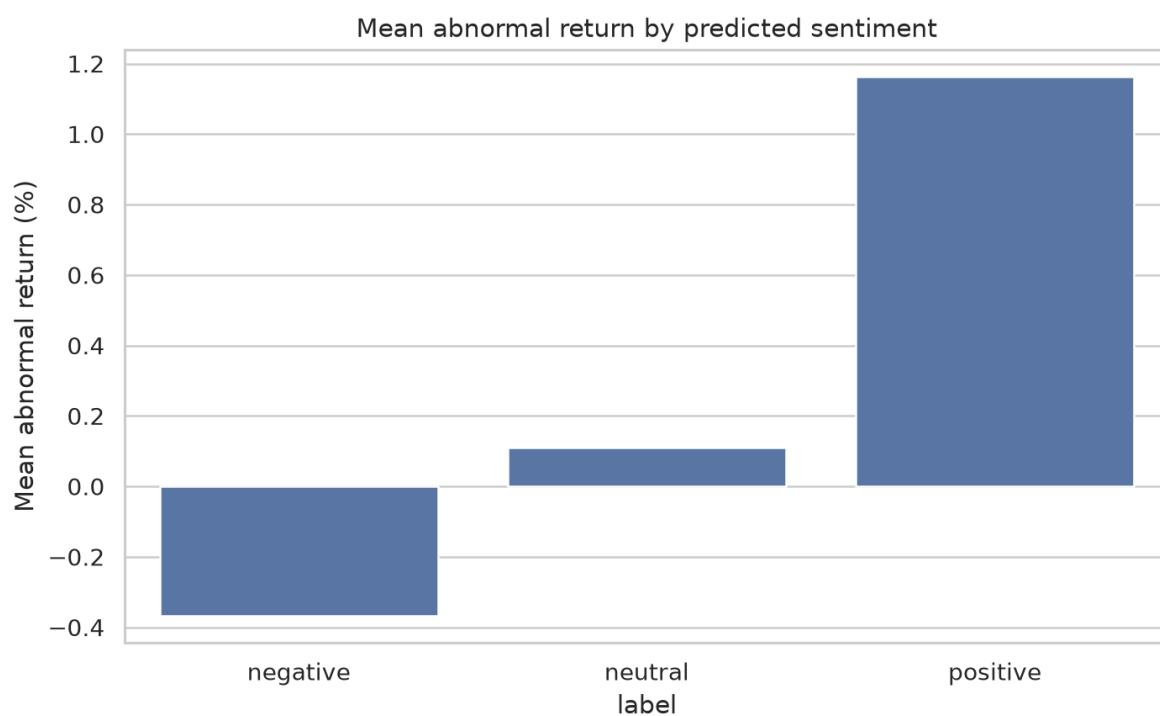


Figure 2. Abnormal return by label

## 7. Interpretation checklist

1. Kiểm tra temporal leakage: bài sau gì cũng có thể không liên quan cho cùng phiên.
2. Kiểm tra entity linking: kết quả chính nên dùng strict/high-confidence mapping.
3. So sánh với baseline và placebo; không chỉ nhìn p-value.
4. Báo cáo effect size và khoảng tin cậy.

5. Nếu intrinsic sentiment tốt nhưng extrinsic không có tín hiệu, đó vẫn là kết quả hợp lệ của mô hình NLP.

## 8. Limitations

CafeF là một nguồn tin, không phải là điểm toàn bộ thông tin thị trường. Timestamp, entity linking, coverage của ticker và của adjusted price cần được kiểm tra thêm công. Kết quả quan sát không chứng minh bài báo gây ra lợi nhuận.

## References

[1] FinABSA-Valid repository: <https://github.com/Khoaph1709/FinABSA-Valid>

[2] CafeF: <https://cafef.vn/>

[3] vnstock: <https://github.com/thinh-vu/vnstock>